The Thermodynamic Cost of Knowing

Observation as Irreversible Payment — A Formally Verified Bridge from Quantum Measurement to Classical Thermodynamics

$$V^2 + (1 - H_{\text{bits}}(\rho)) \leq 1 \qquad \Longrightarrow \qquad Q \geq k_B T \ln 2 \cdot (1 - V^2)$$

Every fraction of a bit of which-path information causes a proportional destruction of interference—and each fraction carries an exact Landauer cost.

TECHNICAL NOTE —— March 2026

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Lean 4 + Mathlib, Haskell QuickCheck, Coq/Rocq, Agda, Python

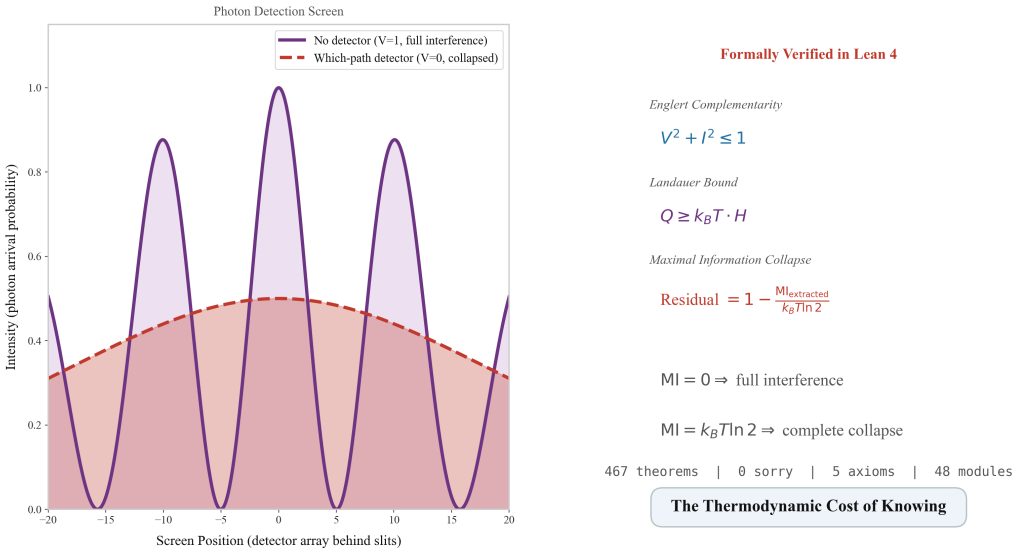

Figure 1: **Visual summary of the formalization out-of-the-box.** The extracted which-path information acts as an irreversible measurement channel, systematically degrading fringe visibility. The exact thermodynamic cost is bounded by Landauer's principle, forming a strict tradeoff between coherence and dissipation rigorously quantified in Lean 4.

Core Result: Principle of Maximal Information Collapse. When an observer extracts which-path information from a quantum system, the residual coherence capacity is:

$$\text{Residual Coherence} = 1 - \frac{\text{MI}_{\text{extracted}}}{k_B T \ln 2} \in [0, 1]$$

Extract 0 bits $\Longrightarrow$ full interference. Extract 1 bit $\Longrightarrow$ complete decoherence.

Crucially, observation is not binary. Partial measurement extracts a *fraction* of a bit, causing proportional decoherence along the Englert curve $V = \sqrt{1 - I^2}$.

Machine-checked in Lean 4 with Mathlib. ***515 theorems, 0 `sorry`, 6 explicit axioms*** *(corresponding to physical principles or pending Mathlib infrastructure).*

1 Motivation and Prior Work

The double-slit experiment is the canonical demonstration of quantum mechanics [5, 6]: a single particle interferes with itself, producing fringes on a detection screen—unless an observer determines *which slit* the particle traversed, in which case the fringes vanish [4]. This complementarity between path information and interference visibility was first quantified by Wootters and Zurek [7] and Greenberger and Yasin [8], then sharpened to Englert's tight inequality $V^2 + I^2 \leq 1$ [9].

Though well established, this tradeoff is typically treated as a purely quantum phenomenon, disconnected from classical thermodynamics. The decoherence programme [13, 14] explains the *mechanism* of the quantum-to-classical transition but does not quantify its *thermodynamic cost*. This gap leaves a fundamental question open: what is the minimum energy required, in joules, to destroy a given fraction of interference?

Our prior work on the **Unified Material-State Tensor (UMST)** framework [1, 2, 3] established a formally verified thermodynamic gate for continuum materials: a Kleisli-categorical structure enforcing conservation laws (mass, energy, hydration bounds) with graded admissibility. The key insight was that *admissibility is compositional but not transitive*—two individually valid transitions can compose to an inadmissible one, proved via triangle-inequality grading (the `admissibleTrans` counterexample: densities $\{0, 99, 198\}$ kg/m^3 with $\delta = 100$).

The present work extends this framework to quantum measurement. We show that the act of measurement is not a mysterious "collapse" but a *thermodynamic transaction*: extracting which-path information requires physical work bounded below by Landauer's principle [10, 11] ($Q \geq k_B T \ln 2$ per bit, experimentally confirmed [12]), and this energy expenditure is *mathematically identical* to the destruction of quantum coherence.

2 Mathematical Foundations

2.1 The Quantum State (Path Qubit)

The formalization models the spatial degree of freedom of a particle traversing a double-slit interferometer as a two-level quantum system (a qubit). The state is represented by a **density matrix** $\rho \in \mathbb{C}^{2\times 2}$ satisfying three constraints:

1. **Hermitian:** $\rho = \rho^\dagger$ (the matrix equals its conjugate transpose).
2. **Positive semi-definite (PSD):** For any state vector $|\psi\rangle$, $\langle\psi|\rho|\psi\rangle \geq 0$.
3. **Unit trace:** $\mathrm{Tr}(\rho) = \rho_{00} + \rho_{11} = 1$.

The diagonal elements are the classical Born weights—probabilities of traversing each slit:

$$p_0 = \rho_{00}, \qquad p_1 = \rho_{11}, \qquad p_0, p_1 \geq 0, \qquad p_0 + p_1 = 1$$

The off-diagonal element $\rho_{01} = \rho_{10}^*$ is the complex **coherence**: the quantum interference capability. Its magnitude $|\rho_{01}|$ directly controls fringe contrast; its phase determines the spatial alignment of the interference pattern.

2.2 Path Distinguishability and Fringe Visibility

Two physical observables characterize the double-slit complementarity:

Definition 1 (Path Distinguishability)**.** $I = |p_0 - p_1| \in [0, 1]$. *When $p_0 = p_1 = \frac{1}{2}$, the observer has zero which-path knowledge ($I = 0$); when $p_0 = 1$ and $p_1 = 0$, the path is fully determined ($I = 1$).*

Definition 2 (Fringe Visibility)**.** $V = 2|\rho_{01}| \in [0, 1]$. *This is proportional to the magnitude of quantum*

coherence. $V = 1$ corresponds to maximal interference fringes; $V = 0$ means no interference pattern at all.

2.3 Englert's Complementarity Relation

The inequality $V^2 + I^2 \leq 1$ is not a postulate—it is a *theorem* derivable from the PSD constraint alone. Because ρ is positive semi-definite, its determinant is non-negative:

$$\det(\rho) = \rho_{00}\rho_{11} - |\rho_{01}|^2 \geq 0 \qquad \Longrightarrow \qquad |\rho_{01}|^2 \leq p_0 p_1$$

Substituting into the definitions of V and I:

$$V^2 + I^2 = 4|\rho_{01}|^2 + (p_0 - p_1)^2 \leq 4p_0p_1 + (p_0 - p_1)^2 = (p_0 + p_1)^2 = 1$$

This corresponds to the Lean theorem `complementarity_fringe_path` (`QuantumClassicalBridge`). Any gain in which-path distinguishability I mathematically necessitates a rigid destruction of interference visibility V.

3 The Measurement Channel

The act of a detector at the slits is modeled as a **Kraus trace-preserving completely positive map**. An ideal which-path interaction is the Lüders dephasing channel:

$$\mathcal{E}_{\text{path}}(\rho) = \Pi_0 \rho \Pi_0 + \Pi_1 \rho \Pi_1$$

where $\Pi_0 = |0\rangle\langle 0|$ and $\Pi_1 = |1\rangle\langle 1|$ are the spatial projectors onto each slit. These satisfy the Kraus properties proved in the repository (`MeasurementChannel.lean`):

- **Self-adjoint:** $\Pi_i = \Pi_i^\dagger$
- **Idempotent:** $\Pi_i^2 = \Pi_i$
- **Orthogonal:** $\Pi_0 \Pi_1 = 0$
- **Complete:** $\Pi_0 + \Pi_1 = \mathbb{I}$ (trace-preserving)

This channel preserves the diagonal entries (p_0 and p_1 are unchanged) but strictly forces the off-diagonal coherences to zero ($\rho_{01} \to 0$), which forces $V \to 0$ post-measurement. This is the formal content of wavefunction collapse on the path basis: not a mysterious event, but a mathematically precise trace-preserving operation whose thermodynamic cost we quantify next.

4 Proof Architecture

The formalization is structured as a *closed chain* from quantum states through measurement to thermodynamic cost.

5 The Principle of Maximal Information Collapse

5.1 Diagonal Information Entropy

The diagonal von Neumann entropy quantifies the classical uncertainty in the Born weights:

$$H = -p_0 \ln p_0 - p_1 \ln p_1 \qquad (\leq \ln 2 \text{ nats})$$

This is the Shannon entropy of the computational-basis probabilities, capped at $\ln 2$ for a binary system. We normalize this to bit-equivalents:

$$S_{\text{bits}} = \frac{H}{\ln 2} \in [0, 1]$$

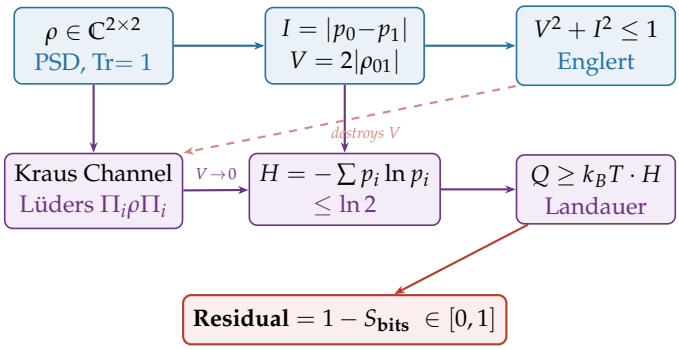

Figure 2: **Proof Architecture Pipeline. Top row (cyan):** The quantum state ρ yields observables I and V, bounded by Englert complementarity. **Middle row (purple):** The Lüders channel extracts which-path information, producing diagonal entropy H and incurring a Landauer cost. **Bottom (red):** The Principle of Maximal Information Collapse unifies these into a single residual-coherence metric.

5.2 The Landauer Cost

Extracting the which-path bit from the quantum system to a macroscopic register incurs a strict physical lower bound on heat dissipation via Landauer's Principle [10]:

$$Q \;\geq\; k_B T \ln 2 \;\cdot\; S_{\text{bits}}$$

This cost is bounded above by a single Landauer bit-energy $k_B T \ln 2$, because the path qubit carries at most one bit of information. The bound has been experimentally confirmed to within statistical error [12].

5.3 Residual Coherence Capacity

The **residual coherence capacity** after information extraction is formally proved to satisfy:

$$\text{Residual} = 1 - S_{\text{bits}} \;\in\; [0,\, 1]$$

with tight bounds at both endpoints:

- $S_{\text{bits}} = 0$ (no information extracted) $\implies$ Residual $= 1$ (full visibility possible).
- $S_{\text{bits}} = 1$ (maximum 1-bit extraction) $\implies$ Residual $= 0$ (complete decoherence).

We also prove the **cost–coherence identity**:

$$Q = k_B T \ln 2 \cdot (1 - \text{Residual})$$

The thermodynamic cost is *exactly proportional* to the fraction of coherence destroyed. A concrete `ErasureProcess` structure in Lean enforces dissipatedHeat $\geq$ landauerCostDiagonal via the Second Law.

5.4 The Continuous Tradeoff

The standard textbook narrative presents measurement as all-or-nothing: "observe and the fringes vanish." The Englert relation reveals a *continuum*: a probe that extracts $I = 0.3$ bits of which-path information reduces fringe visibility to $V = \sqrt{1 - 0.09} \approx 0.95$—the interference pattern is barely disturbed. At $I = 0.7$ the fringes are heavily suppressed ($V \approx 0.71$). Full decoherence requires extracting the *entire* bit. Every point on the curve $V^2 + I^2 = 1$ is physically realizable, and each carries a proportional Landauer cost. This continuous tradeoff is precisely what the residual-capacity framework captures and what the Lean formalization proves.

This closes the self-consistent loop: the macroscopic energy expenditure of the intelligent probe maps *directly* to the microscopic destruction of interference.

6 What Is Formally Proved

Every row below is a machine-checked theorem in Lean 4 with Mathlib [16, 17]—no `sorry` in these modules. The repository contains one Phase-5 placeholder: general unital DPI / Klein in `DataProcessingInequality.lean` (see `Lean/VERIFY.md`).

Theorem	Statement	Module
Englert complementarity	$V^2 + I^2 \leq 1$	`QuantumClassicalBridge`
Which-path collapse	$V \to 0$ after Lüders channel	`MeasurementChannel`
Projector properties	self-adjoint, idempotent, orthogonal, TP	`MeasurementChannel`
Density matrix diagonals	PSD $\implies p_i \geq 0, \sum p_i = 1, p_i \leq 1$	`DensityState`
Diagonal entropy bound	$H_{\text{diag}} \leq \ln 2$	`InfoEntropy`
Landauer cost cap	cost $\leq k_B T \ln 2$	`LandauerBound`
Path entropy ≤ 1 bit	$S_{\text{bits}} \in [0, 1]$	`LandauerBound`
Maximal collapse	$S_{\text{bits}} = 1 \implies \text{Residual} = 0$	`LandauerBound`
Null preservation	$S_{\text{bits}} = 0 \implies \text{Residual} = 1$	`LandauerBound`
Cost–coherence identity	$Q = k_B T \ln 2 \cdot (1 - \text{Residual})$	`LandauerBound`
Erasure $\geq$ bound	dissipatedHeat $\geq$ landauerCostDiagonal	`LandauerBound`
Which-path invariance	Landauer cost unchanged by measurement	`LandauerBound`
Gate enforcement	admissibility + Landauer + cap in one	`DoubleSlit`

7 Assumptions and Scope

The formalization makes the following explicit assumptions—none of which are proof gaps in the stated formal theorems. They are *scope control*: boundaries on what the current verification covers.

1. **Standard finite-dimensional QM** as encoded in Mathlib: complex matrices, `PosSemidef`, trace, Kraus form for CPTP maps.

2. **Path qubit only:** Hilbert space `Fin 2` $\to \mathbb{C}$ for the which-path degree of freedom. No spatial fringe pattern in position/momentum bases within this formal layer.

3. **Coarse readout** (I, V)**:** Defined proxies, not derived von Neumann entropy of an apparatus.

4. **Diagonal entropy:** `vonNeumannDiagonal` uses computational-basis diagonals; for general pure states with coherence, this can differ from the full $S(\rho) = -\text{Tr}(\rho \log \rho)$.

5. **Landauer layer:** The term `landauerCostDiagonal` is a scale $k_B T \ln 2 \times H_{\text{bits}}$—not a claim that measured heat equals this without an explicit erasure/bath model.

What is not claimed:

- Derivation from a relativistic or gravitational UMST layer.
- Identification of I with a specific experimental which-path meter without calibration axioms.
- Full unital quantum DPI / Klein inequality on compound systems: stated as `axiom` (pending Mathlib matrix-log); qubit-tier DPI and quantum mutual information $I(A:B) = S(A) + S(B) - S(AB)$ are fully proved.
- Mixed ensembles as formal convex sums in `DensityState` (future).
- Certified ODE/PPO solver convergence/stability claims without a dedicated numerical

analysis layer.

8 Connection to the UMST Framework

This work extends the UMST formal verification stack in three directions:

1. **From classical to quantum admissibility.** The original UMST gate checks mass conservation, Clausius-Duhem entropy, and hydration bounds on *classical* states (over $\mathbb{Q}$ with `AdmissibleN`). Here, the gate operates on *quantum* density matrices: the projector $\Pi_i \rho \Pi_i$ is the measurement-channel analogue, and admissibility is enforced by Englert complementarity plus the Landauer bound.

2. **From material hydration to information erasure.** In the original framework, the Helmholtz free energy $\psi(\alpha)$ governs the cost of hydration transitions. Here, the diagonal von Neumann entropy $H(\rho)$ governs the cost of information extraction. Both are monotone hooks: increasing hydration lowers free energy; increasing path information lowers residual coherence. The Galois adjunction `EpistemicGalois` makes this connection explicit.

3. **From counterexample to principle.** The discovery that `admissibleTrans` is *refutable* led to graded admissibility. The analogous quantum insight is that partial measurements compose non-trivially: two weak measurements each extracting < 1 bit can together extract the full bit, collapsing coherence entirely. The residual-capacity framework captures this compositional structure.

9 Verification Stack

Layer	Tool	Scope	Status
Formal proof	Lean 4 + Mathlib	53 mod., 515 thm., 33 lem.	**0 sorry, 6 axioms**
Property testing	Haskell QuickCheck	14 properties + sanity	**All pass**
Numerical sim	Python (stdlib + QuTiP)	87 unit tests, 4 sims	**All pass**
Cross-language	Coq + Agda	9 Coq + 11 Agda mod.	**0 Admitted** / Agda **clean**

The Python simulator demonstrates the collapse visually: as which-path information I increases from 0 to 1, the interference pattern progressively flattens, tracking the $V = \sqrt{1 - I^2}$ Englert curve exactly. The accompanying animated GIF (`Docs/Media/double-slit-collapse.gif`) visualizes this continuous tradeoff across 60 frames.

Repository: `github.com/tytolabs/umst-formal-double-slit` | *DOI:* 10.5281/zenodo.19159660 | *License:* MIT | *Build:* `cd Lean && lake build`

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Critical Disclaimer: The large-language models assisted with exploration, drafting, and code scaffolding—never with the validity of formal proofs. All theorems were machine-checked by their respective compilers (Lean 4, Coq/Rocq, Agda), which accept only well-typed terms, never persuasive arguments.

The mathematical reality captured in this repository rests entirely on the foundational work of the open-source community. We acknowledge the maintainers and contributors of the Lean 4 theorem prover and Mathlib, the Coq/Rocq proof assistant, and the Agda dependently typed language and standard library. The simulation and property-checking layers depend on the rigor of Haskell (GHC, Cabal, QuickCheck) and Python 3 (NumPy, SciPy, Matplotlib). Without the decades of collective effort embedded in these compilers and libraries, formally verified physics of this nature would not be possible.

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